

SDS 375 Final Project Written Report

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Research Question and Objective

My research question was to what extent home ice advantage affects NHL games, and to what extent travel, rules, and fans contribute to home ice advantage. I came to this research question as I initially focused on examining the significance of home ice advantage in NHL games, along with all possible factors that could influence it. After getting feedback, I revised my question to focus on a few key variables that I believed would have the greatest impact and were explored through previous historical analysis in the NHL. These variables are travel, which looks at the distance each individual team had to travel, rules, which quantifies how specific rules in hockey favor the home team, and fans, which analyzes the impact of fan attendance for the home team. Additionally, I first started by calculating the impact of home ice advantage on NHL teams in the past five years.

I hypothesized that home teams would gain a measurable advantage in NHL games due to the benefits from rules such as favorable face off positioning, increased travel distance, and time zone changes would negatively affect visiting teams' performance, and that higher crowd attendance and noise levels would work in favor of the home team. What makes my research question different from what others have done before is the fact that I am looking into some unique factors that have not been explored in depth before.

Data Collection and Preparation

I used a total of 4 datasets in the end to complete my project. To analyze whether the distance a team traveled impacted their win percentage, I combined two datasets. I used the NHLscraper package in R to get a dataset that included every individual NHL game. I filtered it to only include regular-season games from 2022 to 2024 and added columns for whether the away team wins and goal differential. Then I obtained a table of the distances in kilometers between every NHL team on https://www.deep13.xyz/nhldistance#distance_table. From this table, I created a dataset with all of the distances and team names, and abbreviations. I joined the distances table to the games table so that it allowed me to carry out a model. The final dataset included the team name, abbreviation, goals for each team, goal differential, if the away team won, location, distance traveled, and other variables relating to the game.

To analyze whether fan attendance impacted teams' win percentage, I looked at data from <https://www.hockey-reference.com/>. This dataset was downloadable in a .csv form, had data from the previous 10 seasons, and had the variables attendance, home team name and score, visiting team name and score, as well as the date this game was played. Next, I created a “win”

variable, which would compare the score of both the home and away team to determine if the game was a win for the home team or not. Lastly, the only NA values were attendance rates after COVID hit, and the site where the dataset was sourced from stated that no people in attendance would default to NA. Therefore, I changed the NA values to zero. Based on these variables, I was then ready to use the Bradley-Terry model on the data in order to figure out the degree to which attendance rates impacted home ice advantage.

The dataset used for the rules analysis was sourced from the website Hockey Elo Ratings (<https://hockeyeloratings.com/players/faceoffratings>). This website provided a csv. file which was put into R Studio in order to be used for data analysis. The data provided by this website was crucial because it allowed me to look at the winning percentage of players who put their stick down first vs. last, and compare the two. The dataset only considers players with either 100 faceoffs taken in the specific season (I looked at the 2024-2025 season), or 300 career faceoffs taken, and gives a dataset of 225 players. This filter is important because it ensures that the sample size of the data is large enough, and that I am only analyzing true centermen.

Exploratory Data Analysis

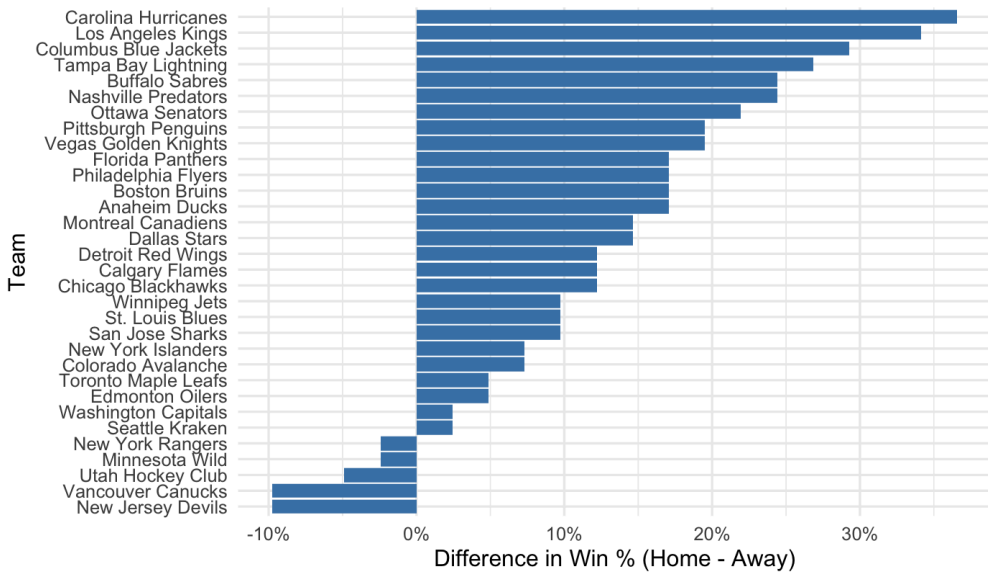
Before investigating which factors have an impact on home ice advantage, I wanted to show evidence that home ice advantage actually does exist in the NHL. To do this, I used the NHLscraper R package and got a dataset of individual games from the past 5 seasons. Then I calculated the winning percentage for home teams and away teams. I found that home teams won 54.2% of the games and away teams won 45.8%. I also wanted to look at it for every team individually, so I created a histogram of the difference in win percentage for home versus away. The graph shows that every team except for 5 had a better win percentage when playing at home versus away. These show that home teams clearly win more than away teams. After seeing this clear relationship, I decided to do some exploratory analysis on whether attendance impacts the home team's win percentage. I compared team win rates in the no-attendance 2020-2021 season to the 2024-2025 season, where fans returned to watch games in person. The correlation between average attendance and change in win rate was $r = -0.11$, indicating no meaningful relationship. Attendance did not appear to have a strong impact on the team's performance. The graph displaying the NHL home ice advantage over time was created by using a Bradley-Terry model, where the intercept found for each season represented the home ice advantage for that season. This was done for two main reasons, the first being that I wanted another proof that home ice advantage was real, and to see if it remained consistent over multiple seasons. I can see that since 2019, the win probability of a home team hovers around 54%.

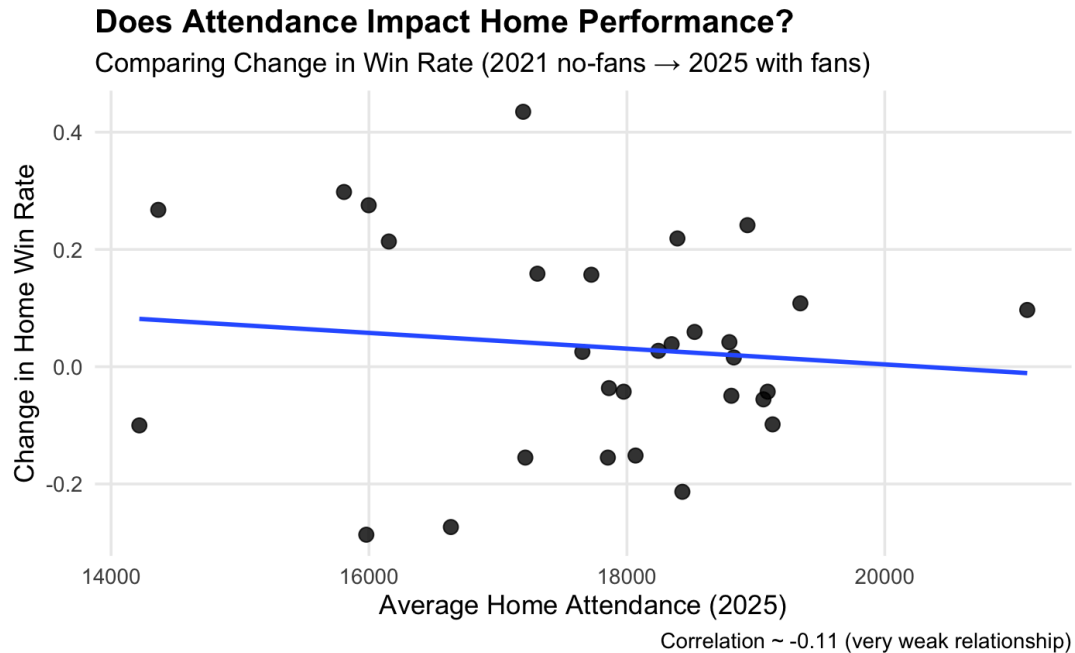
NHL Home Ice Advantage Over Time

Probability a home team wins against an equally-matched opponent



Home Ice Advantage by Team (Home - Away Win%)





Methodology

To conduct the analysis in order to obtain results to make conclusions from, I analyzed three different variables. The variables are fans, travel, and rules. For the fan impact variable, I decided to create a Bradley-Terry model, and this was suggested through the feedback I got from my progress updates. I scraped my data from <https://www.hockey-reference.com/>, and the dataset consisted of attendance rates as well as who the winning team was, so I was able to easily implement the paired data - the change in home win rate and average home attendance between 14,000 to 21,000 fans. From there, I modeled the outcomes using the Bradley-Terry method to determine whether higher attendance provided a measurable advantage in terms of win probability.

For the next variable, travel distance, I chose to create a logistic regression model. I chose this specific model because I wanted to classify the distances into categories in order to understand the patterns better. I scraped my data from the two sources mentioned earlier and then joined the two datasets to create one big one with statistics from individual games and the distance traveled by the away team. I separated the data into three bins for the graph: less than 800km, 800km to 2000km, and more than 2000km of travel. From there, in R, I modeled the probability that the away team wins as a function of the travel distance of the away team. I got the results from that and calculated that the away team's decrease in win percentage for every increase of 1000km was about 1.6%. I also created a graph to show the decrease in win percentage.

To analyze the faceoff data, I took the average of the difference in faceoff winning percentage when going last vs. going first. This gave me a picture of leaguewide, how most players are

impacted by the rule that the home team gets to put their stick down last at center ice faceoffs. Additionally, I created a graph in order to see the amount of players who benefit from putting their stick down last as opposed to first to prove there is an advantage to winning faceoffs that stems from this.

Results

An examination of the past six NHL seasons - 2019/2020 to 2024/2025 - showed that the win probability for home teams averaged at around 54.2%, which demonstrates reasonable evidence for the existence of the effects of home ice advantage.

Using the Hockey Reference dataset, I was able to see the relationship between the number of fans attending each game and home wins. The Bradley-Terry model yielded an intercept of -0.41903, indicating the away team has approximately a 39.7% chance of winning, leaving the home teams' win probability at around 60.3%. While the exploratory data analysis concluded no strong correlation between teams' average fan attendance and change in win rate, the Bradley-Terry model shows that even when two teams are evenly matched, home teams have a measurable advantage in win rate correlated with home attendance numbers.

Call:

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glm(formula = win ~ Home + Visitor, family = binomial(), data = hockey25)
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Coefficients:

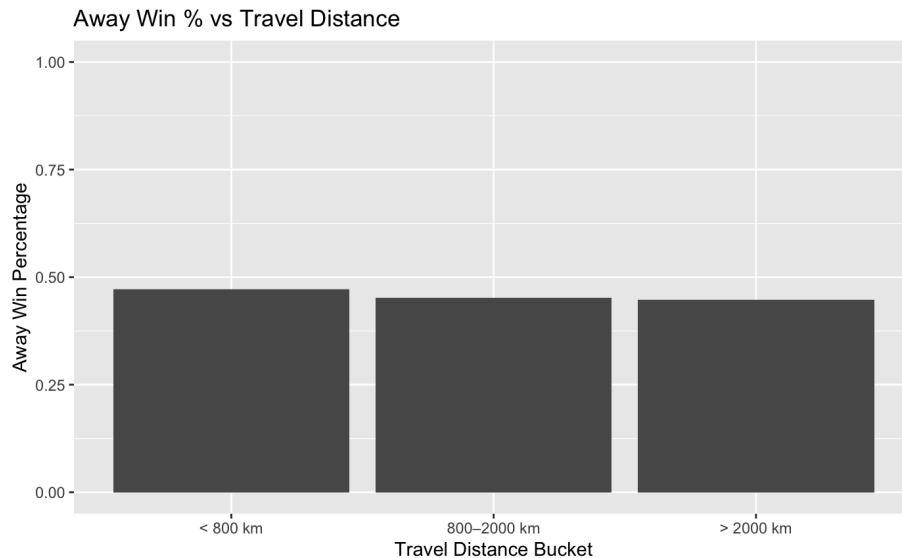
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.41903	0.47356	-0.885	0.37623
HomeBoston Bruins	0.07469	0.45752	0.163	0.87033
HomeBuffalo Sabres	-0.24521	0.45869	-0.535	0.59294
HomeCalgary Flames	-0.17534	0.45937	-0.382	0.70269
HomeCarolina Hurricanes	-1.17974	0.49414	-2.387	0.01697 *
HomeChicago Blackhawks	0.54346	0.46520	1.168	0.24271
HomeColorado Avalanche	-0.55471	0.46561	-1.191	0.23351
HomeColumbus Blue Jackets	-0.55961	0.46534	-1.203	0.22914
HomeDallas Stars	-0.74421	0.47414	-1.570	0.11651

Next, I found through the logistic regression model on travel distance that, while the away teams' baseline win probability lies below 50% and each 1000km increase in travel distance had a slight negative effect on away team win probability, the p-value of 0.592 indicated no statistical significance in this effect. Thus, there is not enough evidence to conclude that travel distance is not a major factor behind home ice advantage in recent NHL seasons.

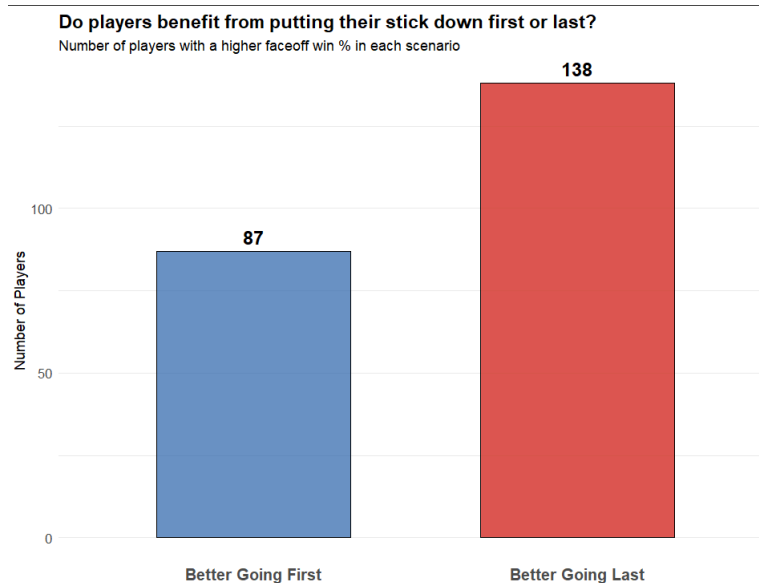
```
Call:
glm(formula = away_win ~ distance_1000km, family = binomial(),
    data = games_travel)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.14716	0.06384	-2.305	0.0212 *
distance_1000km	-0.01585	0.02960	-0.536	0.5923



Finally, my analysis of 225 individual NHL players revealed that putting one's stick down last in faceoffs, which is usually the home team, offers a 1.542% increase in win probability. While this is a small effect on its own, it contributes to the systematic and circumstantial factors that strengthen the presence and effect of home ice advantage, as winning faceoffs increases possession time. It is worth noting that some players do not follow this trend, and perform better putting their stick down first as opposed to last, but for the majority of players (roughly two thirds), putting their stick down last is preferential, and happens more on home ice.



Discussion and Conclusion

After examining the NHL data across the three factors of travel, crowd impact, and rules advantages, I found that home ice advantage is real; however, these factors vary in how they impact home ice advantage.

For the rule-based advantages, I specifically looked at faceoffs, and they were shown (although lightly) to contribute to the home ice advantage. My analysis confirmed that placing the stick down first was disadvantageous, making the home team more likely to win. This supports my hypothesis that NHL rules favor the home team. Next, I looked at fan impact and looked at how COVID (when there were no live audiences for games) impacted games. Using a Bradley-Terry model, there was a clear correlation between attendance rates and home team wins, despite there not appearing to be one in the exploratory data analysis where I just looked at the correlation between attendance and winning percentage. This shows that fan presence boosts home performance and aligns with my initial hypothesis. Lastly, I looked at travel and created a logistic regression model to evaluate wins against the distances the teams had to travel. Travel distances and time zone changes did have an impact on home team wins, but not to the point where it was a significant impact. Therefore, this was different than what I originally thought, and travel distance does not have a significant impact on home ice advantage.

I found that home ice advantage in the NHL does exist, and the rules and fan presence impacted it, while travel was not as significant a factor. This surprised me, given how travel is often cited as one of the key reasons for lower performance in away games, but my analysis says its impact is overestimated. I think this is likely because there have been steady improvements in team travel and recovery over the past decades, and I would be curious to see if travel played a larger

role in the 1980s and 1990s. One limitation that I had was that I wanted to analyze the effects that shootouts had and incorporate that into the rules variable, but I could not find enough data solely on that to analyze it. While datasets often had data on team win percentage in shootouts, they did not have data on whether the team shot first or second, which would be key in quantifying the advantage of the home team.